

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 5

Symmetry and Invariance: Why Some Operations Are Powerful

Parallel Reading

Self-Study Curriculum

Parallel Reading: Hermann Weyl, *Symmetry* (Princeton, 1952): the entire book, which is short (168 pages) and full of visual examples. This is one of the most beautiful books ever written about mathematics for a non-specialist. If unavailable, read: Richard Feynman, *The Character of Physical Law* (1965), Chapter 4 (“Symmetry in Physical Law”), the most accessible treatment of symmetry and conservation ever written, plus PCM section III.94 (Variational Methods).

Also read: Lockhart, *Measurement*, continue Part One (“On Size and Shape”): another ~70 pages. The symmetry arguments inside the shape problems are the reason it is assigned this week.